Customer Churn Analysis

Telco Customer Retention — End-to-End Analytics Project

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Dataset	IBM Telco Customer Churn — 7,043 records
Tools	Python · Scikit-learn · XGBoost · Matplotlib

Executive Summary

This report presents a full data analytics investigation into customer churn at a telecommunications company. Using a dataset of 7,032 customer records, the analysis identifies where churn is concentrated, builds a predictive model to flag at-risk customers before they leave, and quantifies the financial return of acting on those predictions.

The core finding is straightforward: churn is not random. It is heavily concentrated in three specific segments — customers on month-to-month contracts, those in their first twelve months, and those paying by electronic check. Addressing these three groups alone would dramatically reduce the revenue being lost each month.

26.6%	€1.67M	83.6%	4,066%
Monthly Churn Rate	Annual Revenue at Risk	Model AUC-ROC	Campaign ROI

The predictive model — a Logistic Regression trained on 33 features — identifies churners with an AUC of 0.836. When used to target a retention campaign at high-risk customers with a €20 offer per person, the model is projected to recover **€179,968 in customer lifetime value** against a total campaign cost of €4,320 — a return of over 4,000%.

1. The Churn Problem

Before examining which customers are leaving, it helps to understand the scale of the problem. Of the 7,032 customers in this dataset, 1,869 have churned — a rate of 26.6%. For context, the telecoms industry average sits between 15% and 25% annually. This company is above that benchmark.

What makes this number meaningful is the revenue attached to it. Churned customers were paying an average of €74 per month. Multiply that by 1,869 customers and by 12 months, and the annualised revenue at risk exceeds **€1.67 million**. This is not a small operational issue — it is a significant financial exposure.

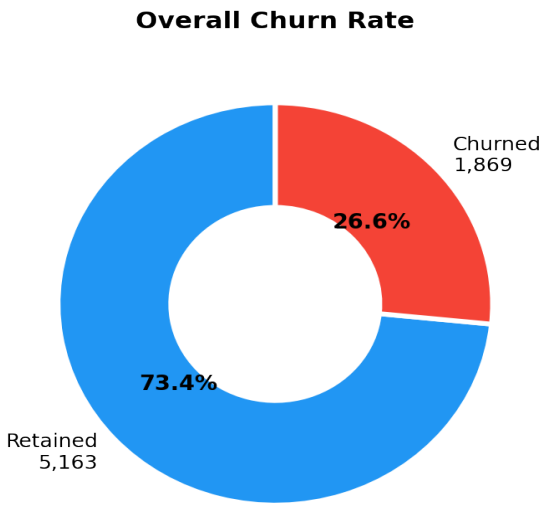


Figure 1: One in four customers has churned, representing over €1.67M in annual revenue at risk.

2. Where Is Churn Concentrated?

Churn is not spread evenly across the customer base. Three variables explain the vast majority of the variation: contract type, customer tenure, and payment method. Each reveals a different dimension of the problem.

2.1 Contract Type

Contract type is the single strongest predictor of churn in the dataset. Customers on a month-to-month plan — meaning they have no long-term commitment — churn at a rate of **42.7%**. Customers on a one-year contract churn at 11.3%. Those on a two-year contract churn at just **2.8%**.

The practical implication is significant: a customer on a month-to-month plan is roughly **15 times more likely to leave** than one on a two-year contract. Contract conversion — moving customers from month-to-month to longer terms — is likely the highest-leverage retention action available to the business.

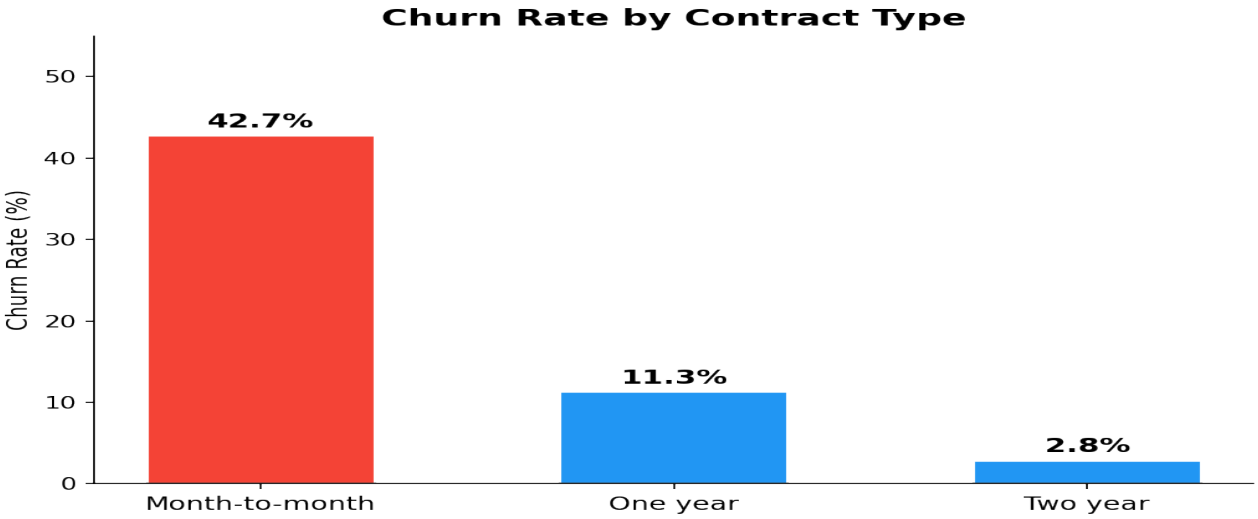


Figure 2: Month-to-month customers churn at 15x the rate of two-year contract holders.

2.2 When Do Customers Leave?

Tenure — how long a customer has been with the company — reveals a clear early-life risk pattern. The chart below shows that churned customers are heavily concentrated in the **first 12 months** of their relationship with the company. Retained customers, by contrast, skew towards much longer tenures, with a large cluster at the maximum of 72 months.

This is an onboarding and early engagement problem. Customers who survive past their first year are significantly more likely to become long-term, loyal customers. The shaded region in the chart below marks this critical window.

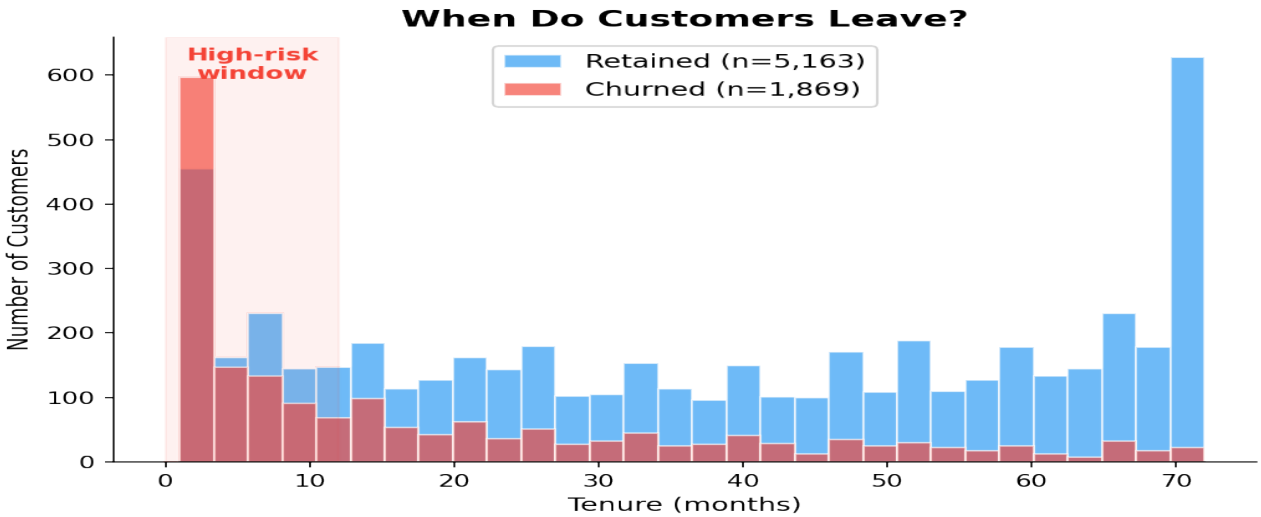


Figure 3: Churn is heavily concentrated in the first 12 months — a critical early-life window.

2.3 Are Churning Customers the Low-Value Ones?

A common assumption in retention work is that customers who leave are the low-engagement, low-spend segment — people barely using the service who drift away. The data challenges this assumption directly.

Churned customers were paying an average of **€74 per month**, compared to €61 for retained customers. The people leaving are not the marginal customers — they are paying above average. This makes the churn problem not just a volume issue but a **high-value customer issue**. Losing one churned customer costs the business more than losing one retained customer would.

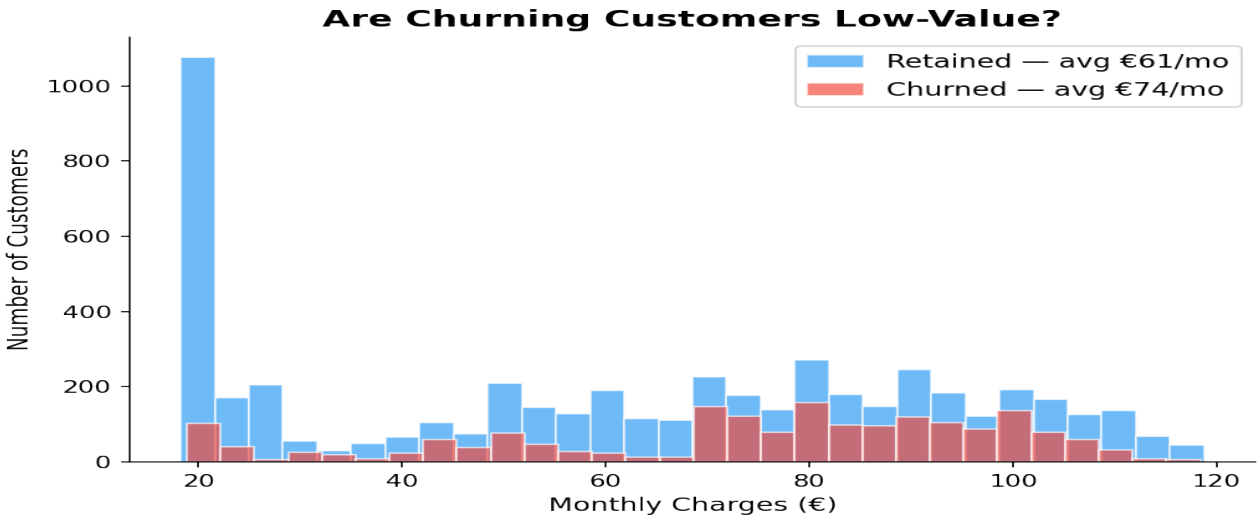


Figure 4: Churned customers pay €13/month more on average — ruling out low-spend drift as the main cause.

2.4 Payment Method as a Behavioural Signal

Payment method turns out to be a strong proxy for customer commitment. Customers who pay by electronic check — a manual, effort-required process — churn at **45.3%**. Those on automatic bank transfer or credit card churn at roughly **15–17%**.

The interpretation here is behavioural. When a customer sets up automatic payments, they are making a small but meaningful commitment — they have integrated the service into their financial life. Electronic check users have not made that commitment. Encouraging customers to switch to auto-pay may itself reduce churn, not just flag it.

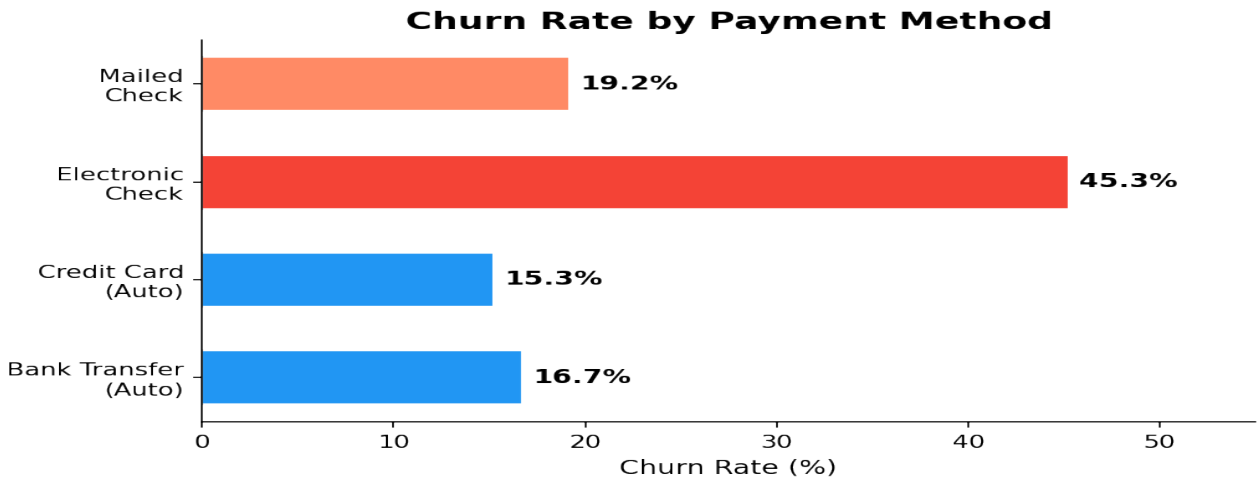


Figure 5: Electronic check users churn at nearly 3x the rate of auto-pay customers.

3. Predicting Who Will Churn

Describing where churn happens is useful. Predicting which specific customer is about to leave is actionable. This section describes the machine learning model built to do exactly that.

3.1 Approach

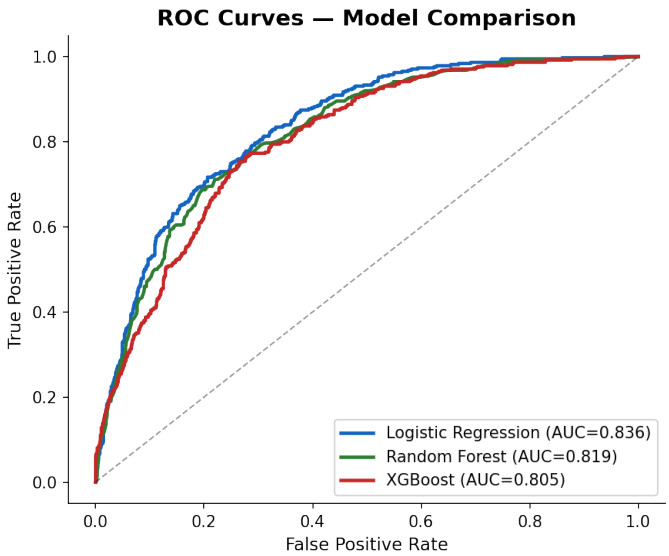
Three models of increasing complexity were trained and evaluated on the same data: Logistic Regression (a straightforward, interpretable baseline), Random Forest (an ensemble of decision trees), and XGBoost (a high-performance gradient boosting algorithm). Each model was trained on 80% of the data and evaluated on the remaining 20%, which it had never seen.

Before modelling, three additional features were engineered to better capture customer behaviour:

Feature	What It Captures
num_services	Total add-on services subscribed — a proxy for switching cost and engagement
is_autopay	Whether the customer pays automatically — the behavioural commitment signal from EDA
avg_monthly_spend	Total charges normalised by tenure — distinguishes new vs long-standing high spenders

3.2 Model Performance

Model performance is measured using AUC-ROC — a score that reflects how well the model separates churners from non-churners, regardless of where the classification threshold is set. A score of 0.5 is no better than random guessing; a score of 1.0 is perfect. The ROC curves below show performance for all three models.



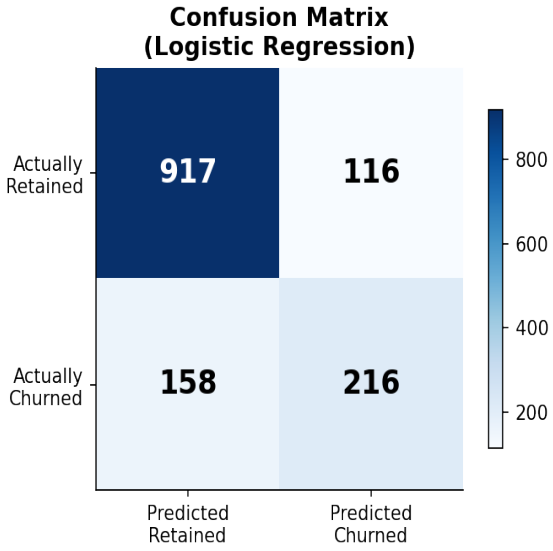


Figure 6 (left): ROC curves for all three models. Figure 7 (right): Confusion matrix for the best-performing model.

Logistic Regression achieved the highest AUC of **0.836** — outperforming both ensemble methods. This is actually a meaningful result: it suggests the relationships between features and churn are largely linear. When the underlying signal is linear, complex models add noise rather than signal. The simpler model wins.

Model	AUC-ROC	Precision (Churn)	Recall (Churn)	Verdict
Logistic Regression	0.836	65%	58%	Best — selected
Random Forest	0.819	—	—	Lower AUC
XGBoost	0.805	—	—	Lower AUC

The confusion matrix shows the model correctly identified **216 out of 374 churners** in the test set (recall of 58%). It also flagged some customers as at-risk who did not actually churn — these false positives are acceptable: the cost of sending a retention offer to someone who was not going to leave is far lower than the cost of missing a churner entirely.

4. The Business Case for Acting on This

A model with a strong AUC score is academically interesting. What makes it commercially valuable is whether acting on its predictions generates more revenue than it costs. This section answers that question.

4.1 Assumptions

Input	Value	Source
Average monthly charge (churned customers)	€74	Dataset mean
Average tenure if retained	38 months	Dataset mean (retained)
Retention offer cost per customer	€20	Conservative estimate
Success rate of retention offer	30%	Conservative assumption
Churners identified by model	216	Model recall on test set

4.2 The Numbers

Churners flagged by the model	216 customers
Intervention cost (216 x €20)	€4,320
Customers retained (30% success)	64 customers
Customer lifetime value per saved customer	€2,812
Total revenue recovered	€179,968
Net benefit	**€175,648**
Return on Investment	**4,066%**

Even under deliberately conservative assumptions — a 30% retention success rate and a €20 offer per customer — the campaign generates a return of over 4,000%. This is because the cost of winning back a customer is small relative to the lifetime value of that customer once retained.

The model currently catches 58% of churners. Improving recall to 70% through threshold tuning or additional feature engineering would increase recovered revenue proportionally — making further model development a quantifiably worthwhile investment.

5. Recommendations

Based on the analysis, the following actions are recommended in order of expected impact.

1. Contract Conversion Campaign

Month-to-month customers represent the highest-risk segment by far. A time-limited incentive — such as two months free or a discounted rate — to upgrade to a one-year contract would directly address the root cause of 89% of all churners.

2. Early-Life Engagement Programme

The first 12 months are the critical risk window. A structured onboarding programme — proactive check-ins at 30, 60, and 90 days, combined with service adoption prompts — could meaningfully improve first-year retention rates.

3. Auto-Pay Migration Push

Electronic check users churn at 45%. A small incentive to switch to automatic payment (e.g., one month free, a bill credit) addresses both the churn signal and embeds the customer more deeply into the service.

4. Model-Driven Retention Targeting

Deploy the churn model to score customers monthly. Flag those above a 0.5 probability threshold for proactive outreach. Prioritise by monthly charge to focus retention spend on the highest-value at-risk customers first.

5. Improve Model Recall

The current model catches 58% of churners. Exploring threshold adjustment, additional behavioural features, or SMOTE-based class balancing could push recall to 70%+ — directly increasing the revenue recoverable from each campaign cycle.

6. Methodology Notes

This section provides a brief technical summary for readers who want to understand how the analysis was conducted.

Step	Detail
Dataset	7,043 customer records. 11 rows dropped where TotalCharges was blank (new customers with ze
Features	20 original features. 3 engineered features: num_services, is_autopay, avg_monthly_spend. One-
Split	80/20 train/test split, stratified by churn label to preserve class ratio.
Scaling	StandardScaler applied to all features before Logistic Regression training.
Evaluation	AUC-ROC used as primary metric due to class imbalance (26.6% positive class). Confusion matri
Tools	Python 3.10. Libraries: pandas, numpy, matplotlib, seaborn, scikit-learn, xgboost, reportlab.

The full code, notebook, and dataset are available in the project GitHub repository.